WoSM Academy

MAT231-F27 · CALCULUS II

The gradient

Directional derivatives and level sets

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WoSM Academy

Lecture 11 · Unit 11 · Fall 2027

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Last time

- Partial derivatives $\frac{\partial f}{\partial x_i}$: differentiate in one variable, freeze the others.
- f is **differentiable** at $\mathbf{a}$ if $f(\mathbf{a}+\mathbf{h}) = f(\mathbf{a}) + Df(\mathbf{a})\,\mathbf{h} + o(\|\mathbf{h}\|)$ for a linear map $Df(\mathbf{a})$.
- Then $Df(\mathbf{a})$ is the matrix of partial derivatives; for $f\colon \mathbb{R}^2 \to \mathbb{R}$ it gives the tangent plane.
- Continuous partial derivatives (C^1) imply differentiability.

Today: for a real-valued f, the row $Df(\mathbf{a})$ becomes a *vector*, the gradient. It collects all first-order information about f near $\mathbf{a}$, and it points uphill.

This lecture

01 Directional derivatives

02 The gradient vector

03 Level sets

04 Reading contour plots

05 Rules for the gradient

06 Application: least squares

07 Summary and practice

Reading: G. Strang, E. Herman, *Calculus, Volume 3* (OpenStax), Section 4.6.

01

Directional derivatives

The rate of change in a direction

Definition

Let $f : U \to \mathbb{R}$, $U \subseteq \mathbb{R}^n$ open, $\mathbf{a} \in U$, and $\mathbf{v}$ a **unit** vector. The **directional derivative** of f at $\mathbf{a}$ in the direction $\mathbf{v}$ is

$$D_{\mathbf{v}} f(\mathbf{a}) = \frac{\mathrm{d}}{\mathrm{d}t} f(\mathbf{a} + t\mathbf{v})\Big|_{t=0} = \lim_{t \to 0} \frac{f(\mathbf{a} + t\mathbf{v}) - f(\mathbf{a})}{t},$$

if the limit exists.

The rate of change in a direction

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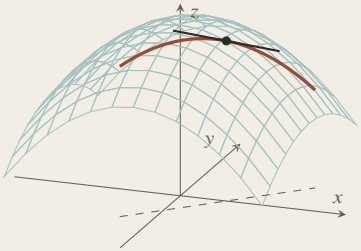
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if the limit exists.

- Walk from $\mathbf{a}$ along the line through $\mathbf{a}$ in direction $\mathbf{v}$: $D_{\mathbf{v}} f(\mathbf{a})$ is how fast f changes, per unit of distance.
- The partial derivatives are the special cases $\mathbf{v} = \mathbf{e}_i$: $D_{\mathbf{e}_i} f = \frac{\partial f}{\partial x_i}$.
- $\mathbf{v}$ must be a unit vector; otherwise the rate is scaled by $\|\mathbf{v}\|$.

A picture



$f(x,y) = 4 - x^2 - \frac{1}{2}y^2$, $\mathbf{a} = (\frac{1}{2}, 0)$, $\mathbf{v} = \frac{1}{\sqrt{2}}(1,1)$.

- The dashed line is $\mathbf{a} + t\mathbf{v}$ in the plane.
- Above it, the graph gives the curve $t \mapsto f(\mathbf{a} + t\mathbf{v})$.
- $D_{\mathbf{v}} f(\mathbf{a})$ is the slope of this curve at $t = 0$: the slope of the graph in the direction $\mathbf{v}$. Here it is $-\frac{1}{\sqrt{2}}$.

Directional derivatives from the derivative

Theorem

If f is differentiable at $\mathbf{a}$, then for every unit vector $\mathbf{v}$: $D_{\mathbf{v}} f(\mathbf{a}) = Df(\mathbf{a})\,\mathbf{v} = \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(\mathbf{a})\, v_i$.

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Proof.

With $\mathbf{h} = t\mathbf{v}$ and $\|t\mathbf{v}\| = |t|$: $f(\mathbf{a} + t\mathbf{v}) = f(\mathbf{a}) + t\,Df(\mathbf{a})\,\mathbf{v} + o(|t|)$. Subtract $f(\mathbf{a})$, divide by t, let $t \to 0$. □

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So one vector, $Df(\mathbf{a})$, gives the rate of change in every direction.

Directional derivatives are not enough

> **Warning**
>
> The converse fails. For $f = \frac{x^2 y}{x^4 + y^2}$, $f(0,0) = 0$ (Lecture 9), every directional derivative at $(0,0)$ exists: $D_{\mathbf{v}} f = \frac{v_1^2}{v_2}$ for $v_2 \neq 0$, and 0 otherwise. Yet f is not even continuous there — and $\mathbf{v} \mapsto D_{\mathbf{v}} f$ is not linear.

As with partial derivatives (Lecture 10): knowing f along every line through $\mathbf{a}$ does not control f near $\mathbf{a}$. Differentiability is the condition that does.

Examples

Example

$f(x,y) = x^2y + y^3$ at $(1,2)$, in the direction of $(3,4)$. The unit vector is $\mathbf{v} = \frac{1}{5}(3,4)$; $f_x = 2xy = 4$, $f_y = x^2 + 3y^2 = 13$; so $D_{\mathbf{v}} f(1,2) = 4 \cdot \frac{3}{5} + 13 \cdot \frac{4}{5} = \frac{64}{5} = 12.8$.

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Example

$f(x,y,z) = xyz$ at $(1,1,1)$, in the direction $\frac{1}{\sqrt{3}}(1,1,1)$: all partial derivatives are 1, so $D_{\mathbf{v}}f = \frac{3}{\sqrt{3}} = \sqrt{3}$.

Interpretation: walking from $(1,2)$ towards $(4,6)$, the graph of f rises 12.8 units per unit of horizontal distance.

02

The gradient vector

The gradient

Definition

The **gradient** of f at $\mathbf{a}$ is the column vector of partial derivatives, $\nabla f(\mathbf{a}) = \left(\frac{\partial f}{\partial x_1}(\mathbf{a}), \ldots, \frac{\partial f}{\partial x_n}(\mathbf{a})\right)^\top$.

- $Df(\mathbf{a}) = \nabla f(\mathbf{a})^\top$: the derivative is a row, the gradient its transpose.
- So for differentiable f: $D_{\mathbf{v}} f(\mathbf{a}) = \nabla f(\mathbf{a}) \cdot \mathbf{v}$, and $f(\mathbf{a}+\mathbf{h}) = f(\mathbf{a}) + \nabla f(\mathbf{a}) \cdot \mathbf{h} + o(\|\mathbf{h}\|)$.

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Example

$\nabla(x^2 + y^2) = (2x, 2y), \quad \nabla e^{xy} = (y\,e^{xy},\ x\,e^{xy}), \quad \nabla(xyz) = (yz, xz, xy).$

The direction of steepest ascent

Theorem

Let f be differentiable at $\mathbf{a}$ with $\nabla f(\mathbf{a}) \neq \mathbf{0}$. Among all unit vectors $\mathbf{v}$, $D_{\mathbf{v}} f(\mathbf{a})$ is largest for $\mathbf{v} = \frac{\nabla f(\mathbf{a})}{\|\nabla f(\mathbf{a})\|}$, where it equals $\|\nabla f(\mathbf{a})\|$, and smallest in the opposite direction.

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Proof.

By the Cauchy–Schwarz inequality (Linear Algebra, Unit 13), $|D_{\mathbf{v}} f(\mathbf{a})| = |\nabla f(\mathbf{a}) \cdot \mathbf{v}| \leq \|\nabla f(\mathbf{a})\|\,\|\mathbf{v}\| = \|\nabla f(\mathbf{a})\|$, with equality exactly when $\mathbf{v}$ is a multiple of $\nabla f(\mathbf{a})$. □

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Equivalently $D_{\mathbf{v}} f(\mathbf{a}) = \|\nabla f(\mathbf{a})\| \cos\theta$, with θ the angle between $\mathbf{v}$ and $\nabla f(\mathbf{a})$; it is 0 perpendicular to $\nabla f(\mathbf{a})$.

Steepest descent

Example (A hiker)

The height of a hill is $f(x, y) = 100 - x^2 - 2y^2$. At $(3, 2)$: $\nabla f = (-2x, -4y) = (-6, -8)$, with $\|\nabla f\| = 10$. Steepest way up: the direction $(-0.6, -0.8)$, at slope 10. Steepest way down: $(0.6, 0.8)$, at slope -10. Level walking: $\pm(0.8, -0.6)$, perpendicular to ∇f.

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Gradient descent (Lecture 15)

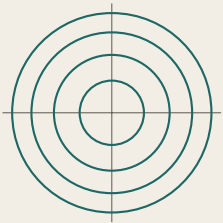
Step against the gradient: $\mathbf{x}_{k+1} = \mathbf{x}_k - \eta\, \nabla f(\mathbf{x}_k)$, with a small step size $\eta > 0$. This simple rule trains most machine-learning models.

03

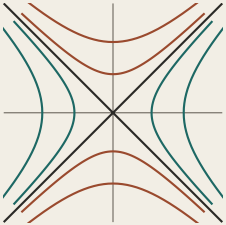
Level sets

Level curves and level surfaces

The **level set** of f at height c is $\{\mathbf{x} : f(\mathbf{x}) = c\}$: a **level curve** for $n = 2$, a **level surface** for $n = 3$. A **contour plot** draws several of them.



$x^2 + y^2$: circles



$x^2 - y^2$: hyperbolas

($c > 0$, $c < 0$, lines for $c = 0$)



$x^2 + 4y^2$: ellipses

Level surfaces in $\mathbb{R}^3$: $x^2 + y^2 + z^2 = c$ are spheres; $x + 2y + 3z = c$ are parallel planes.

The gradient is perpendicular to level sets

Theorem

Let f be differentiable, and $\mathbf{r}(t)$ a differentiable curve in the level set $\{f = c\}$ with $\mathbf{r}(0) = \mathbf{a}$. Then $\nabla f(\mathbf{a}) \cdot \mathbf{r}'(0) = 0$.

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Proof.

$f(\mathbf{r}(s)) = c$ for all s, and by differentiability $0 = f(\mathbf{r}(s)) - f(\mathbf{a}) = \nabla f(\mathbf{a}) \cdot (\mathbf{r}(s) - \mathbf{a}) + o(\|\mathbf{r}(s) - \mathbf{a}\|)$. Divide by s, let $s \to 0$ ($\frac{\|\mathbf{r}(s)-\mathbf{a}\|}{|s|}$ stays bounded): $0 = \nabla f(\mathbf{a}) \cdot \mathbf{r}'(0)$. A first case of the chain rule (Lecture 12). □

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So $\nabla f(\mathbf{a})$ is perpendicular to every curve in the level set through $\mathbf{a}$: it is a **normal vector** to the level set.

Tangent lines and tangent planes

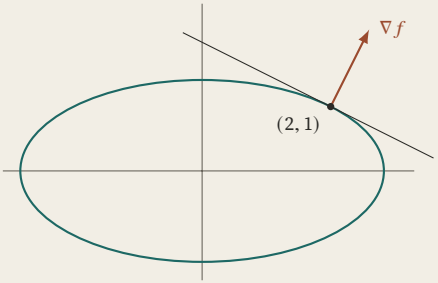
- Level curve, $\nabla f(\mathbf{a}) \neq \mathbf{0}$: the tangent line at $\mathbf{a}$ is $\nabla f(\mathbf{a}) \cdot (\mathbf{x} - \mathbf{a}) = 0$.
- Level surface: the tangent plane at $\mathbf{a}$ is $\nabla f(\mathbf{a}) \cdot (\mathbf{x} - \mathbf{a}) = 0$.

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Example (Ellipse $x^2 + 4y^2 = 8$ at $(2, 1)$)

$\nabla f = (2x, 8y) = (4, 8)$: $4(x - 2) + 8(y - 1) = 0$, that is $x + 2y = 4$.



Tangent planes to level surfaces

Example (Sphere $x^2 + y^2 + z^2 = 9$ at $(1, 2, 2)$)

$\nabla f = (2x, 2y, 2z) = (2, 4, 4)$: $2(x-1) + 4(y-2) + 4(z-2) = 0$, that is $x + 2y + 2z = 9$. The normal points away from the centre, as it should.

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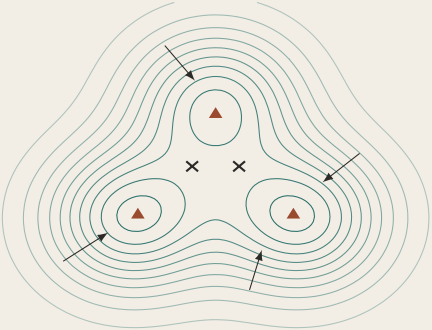
Example (Graphs are level sets)

The graph $z = g(x, y)$ is the level set $F = 0$ of $F(x, y, z) = g(x, y) - z$, with $\nabla F = (g_x, g_y, -1)$. The tangent plane $g_x(a, b)(x - a) + g_y(a, b)(y - b) - (z - g(a, b)) = 0$ is the one from Lecture 10.

04

Reading contour plots

Reading a contour plot



Three hills; levels $0.1, 0.2, \ldots, 1$.

- The gradient (arrows) crosses the contours at right angles, pointing uphill.
- Where contours are close together, $\|\nabla f\|$ is large: the slope is steep.
- $\nabla f = \mathbf{0}$ at the three summits (▲) and at two saddle points (×): the passes between hills.

Topographic maps work the same way: the contours are lines of equal height.

Where the gradient vanishes

- At a summit, every direction goes down: $D_{\mathbf{v}} f(\mathbf{a}) \le 0$ for all $\mathbf{v}$, and also for $-\mathbf{v}$. So $\nabla f(\mathbf{a}) \cdot \mathbf{v} = 0$ for all $\mathbf{v}$, that is $\nabla f(\mathbf{a}) = \mathbf{0}$.
- At a saddle point (a mountain pass), the gradient is also $\mathbf{0}$, but the function goes up in some directions and down in others.
- Near a point where $\nabla f \neq \mathbf{0}$, the contours look like nearly parallel, evenly spaced lines.

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> **Remark**
>
> Points with $\nabla f = \mathbf{0}$ are called **critical points**. Telling summits, pits and saddles apart needs second derivatives: Lectures 13 and 14.

05

Rules for the gradient

Rules

For differentiable f, g and constants α, β:

$$\nabla(\alpha f + \beta g) = \alpha \nabla f + \beta \nabla g, \qquad \nabla(fg) = f\,\nabla g + g\,\nabla f,$$

$$\nabla\left(\frac{f}{g}\right) = \frac{g\,\nabla f - f\,\nabla g}{g^2} \quad (g \neq 0), \qquad \nabla\big(\varphi(f)\big) = \varphi'(f)\,\nabla f.$$

Each follows from the one-variable rule applied to each partial derivative. The last is a case of the chain rule (Lecture 12).

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> **Example**
>
> $\nabla\|\mathbf{x}\| = \nabla\sqrt{\|\mathbf{x}\|^2} = \frac{1}{2\|\mathbf{x}\|}\nabla\|\mathbf{x}\|^2 = \frac{\mathbf{x}}{\|\mathbf{x}\|}$ for $\mathbf{x} \neq \mathbf{0}$: a unit vector pointing away from $\mathbf{0}$.

Gradients in vector notation

For $\mathbf{x} \in \mathbb{R}^n$, a fixed vector $\mathbf{a}$ and a fixed $n \times n$ matrix A:

$$\nabla(\mathbf{a} \cdot \mathbf{x}) = \mathbf{a}, \qquad \nabla\left(\|\mathbf{x}\|^2\right) = 2\mathbf{x}, \qquad \nabla\left(\mathbf{x}^\top A \mathbf{x}\right) = (A + A^\top)\mathbf{x}.$$

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Proof of the last one.

$\mathbf{x}^{\top} A \mathbf{x} = \sum_{i,j} a_{ij} x_i x_j$. Differentiating in x_k, the terms with $i = k$ give $\sum_j a_{kj} x_j = (A\mathbf{x})_k$ and those with $j = k$ give $\sum_i a_{ik} x_i = (A^{\top}\mathbf{x})_k$. □

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- For symmetric A: $\nabla(\mathbf{x}^{\top} A \mathbf{x}) = 2A\mathbf{x}$ — like $(ax^2)' = 2ax$.
- These rules are used constantly in machine learning: they turn “differentiate a loss” into a few lines of matrix algebra.

06

Application: least squares

The normal equations

Given an $m \times n$ matrix A and $\mathbf{b} \in \mathbb{R}^m$, find $\mathbf{x}$ minimising $E(\mathbf{x}) = \|A\mathbf{x} - \mathbf{b}\|^2$.

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$$E(\mathbf{x}) = (A\mathbf{x} - \mathbf{b})^\top (A\mathbf{x} - \mathbf{b}) = \mathbf{x}^\top A^\top A\mathbf{x} - 2\,(A^\top \mathbf{b}) \cdot \mathbf{x} + \mathbf{b}^\top \mathbf{b}.$$

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At a minimum the gradient is $\mathbf{0}$:

$$A^\top A\,\mathbf{x} = A^\top \mathbf{b} \qquad \text{(the normal equations)}.$$

That every solution is a minimum follows from the second derivative $2A^\top A$ (Lectures 13–14).

Two routes to one equation

- **Linear Algebra** (Units 14 and 16): $A\mathbf{x}$ is closest to $\mathbf{b}$ when it is the orthogonal projection of $\mathbf{b}$ onto the column space of A. Then the residual $\mathbf{b} - A\mathbf{x}$ is perpendicular to every column: $A^{\top}(\mathbf{b} - A\mathbf{x}) = \mathbf{0}$.
- **Calculus:** set the gradient of $\|A\mathbf{x} - \mathbf{b}\|^2$ to zero.

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The same equation $A^\top A\mathbf{x} = A^\top\mathbf{b}$. Geometry explains *why* the answer is a projection; calculus generalises to losses that are not quadratic, where no projection exists — and there we use gradient descent.

Example: fitting a line

Fit $y = c_0 + c_1 t$ to the points $(0,1)$, $(1,2)$, $(2,2)$:

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix},$$

$$A^{\top}A = \begin{pmatrix} 3 & 3 \\ 3 & 5 \end{pmatrix}, \quad A^{\top}\mathbf{b} = \begin{pmatrix} 5 \\ 6 \end{pmatrix}.$$

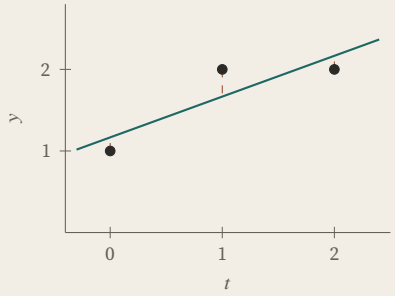
Example: fitting a line

Fit $y = c_0 + c_1 t$ to the points $(0,1)$, $(1,2)$, $(2,2)$:

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix},$$

$$A^{\top}A = \begin{pmatrix} 3 & 3 \\ 3 & 5 \end{pmatrix}, \quad A^{\top}\mathbf{b} = \begin{pmatrix} 5 \\ 6 \end{pmatrix}.$$

Solving: $c_1 = \frac{1}{2}$, $c_0 = \frac{7}{6}$. The line $y = \frac{7}{6} + \frac{t}{2}$ minimises the sum of the squared vertical distances.



07

Summary and practice

Summary

- **Directional derivative**: $D_{\mathbf{v}} f(\mathbf{a}) = \frac{\mathrm{d}}{\mathrm{d}t} f(\mathbf{a} + t\mathbf{v})|_{t=0}$, for a unit vector $\mathbf{v}$; the slope of the graph in direction $\mathbf{v}$.
- **Gradient** ∇f: the vector of partial derivatives. For differentiable f, $D_{\mathbf{v}} f = \nabla f \cdot \mathbf{v}$.
- ∇f points in the direction of steepest ascent, with rate $\|\nabla f\|$; $-\nabla f$ is steepest descent (gradient descent).
- ∇f is perpendicular to level sets; tangent lines and planes: $\nabla f(\mathbf{a}) \cdot (\mathbf{x} - \mathbf{a}) = 0$.
- Rules: $\nabla(\mathbf{a} \cdot \mathbf{x}) = \mathbf{a}$, $\nabla\|\mathbf{x}\|^2 = 2\mathbf{x}$, $\nabla(\mathbf{x}^\top A\mathbf{x}) = (A + A^\top)\mathbf{x}$.
- Least squares: $\nabla\|A\mathbf{x} - \mathbf{b}\|^2 = 0$ gives $A^\top A\mathbf{x} = A^\top \mathbf{b}$.

Practice

1. Find $D_{\mathbf{v}} f$ for $f(x,y) = xe^{y}$ at $(2,0)$ in the direction of $(1,-1)$.
2. Find the direction of steepest ascent of $f(x,y,z) = x^2 y + yz^2$ at $(1,1,1)$, and the rate of increase in that direction.
3. Find the tangent line to the curve $x^3 + y^3 = 9$ at $(1,2)$.
4. Find the tangent plane to the ellipsoid $x^2 + 2y^2 + 3z^2 = 6$ at $(1,1,1)$.
5. Show that $\nabla \frac{1}{\|\mathbf{x}\|} = -\frac{\mathbf{x}}{\|\mathbf{x}\|^3}$ for $\mathbf{x} \neq \mathbf{0}$.
6. For $A = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$, compute $\nabla(\mathbf{x}^{\top} A \mathbf{x})$ directly and with the formula.
7. Fit a line $y = c_0 + c_1 t$ to $(0,0)$, $(1,1)$, $(2,1)$, $(3,2)$ by least squares.
8. On a contour plot, in which directions is $D_{\mathbf{v}} f(\mathbf{a}) = 0$? Explain.

Answers

1. $\nabla f = (e^y, xe^y) = (1,2)$, $\mathbf{v} = \frac{1}{\sqrt{2}}(1,-1)$: $D_{\mathbf{v}} f = -\frac{1}{\sqrt{2}}$.
2. $\nabla f = (2xy,\ x^2+z^2,\ 2yz) = (2,2,2)$: direction $\frac{1}{\sqrt{3}}(1,1,1)$, rate $2\sqrt{3}$.
3. $\nabla = (3x^2, 3y^2) = (3,12)$: $3(x-1)+12(y-2)=0$, $x+4y=9$.
4. $\nabla = (2,4,6)$: $x+2y+3z=6$.
5. $\nabla(\|\mathbf{x}\|^2)^{-1/2} = -\frac{1}{2}(\|\mathbf{x}\|^2)^{-3/2}\cdot 2\mathbf{x}$.
6. $\mathbf{x}^\top A\mathbf{x} = x_1^2 + 2x_1x_2 + 3x_2^2$, so $\nabla = (2x_1+2x_2,\ 2x_1+6x_2)$; and $A + A^\top = \begin{pmatrix} 2 & 2 \\ 2 & 6 \end{pmatrix}$.
7. $A^\top A = \begin{pmatrix} 4 & 6 \\ 6 & 14 \end{pmatrix}$, $A^\top \mathbf{b} = \begin{pmatrix} 4 \\ 9 \end{pmatrix}$: $y = 0.1 + 0.6\,t$.
8. Along the contour through $\mathbf{a}$ (perpendicular to $\nabla f(\mathbf{a})$): there f is momentarily constant.

Next time

Lecture 12 — The chain rule (Unit 12)

- The derivative of a composition is the product of the Jacobian matrices
- Functions along curves, change of variables, implicit differentiation
- Computational graphs and backpropagation; the Jacobian determinant

Reading: Strang & Herman, *Calculus, Volume 3*, Section 4.5.

Questions: the module channel on Discord, or office hours, Monday–Friday, 10:00–17:00 (Baku time, UTC+4), ali.khudiyev@wosm.academy.